

STAT 423 Winter 2026 - Final Project

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```
orig_data <- read.csv("billboard.csv")

cleaned_data <- data.frame(artist = orig_data$artist, rating=orig_data$overall_rating, age=orig_data$fr

cleaned_data$gender <- ifelse(orig_data$artist_male == 1, "Male", "Not Male")

cleaned_data$race <- ifelse(orig_data$artist_white == 1, "White", "Not White")

cleaned_data$genre <- ifelse(is.na(orig_data$cdr_genre), "Genre Not Specified", orig_data$cdr_genre)

cleaned_data$genre <- ifelse(grepl("Pop", orig_data$cdr_genre), "Pop",
                             ifelse(grepl("Country", orig_data$cdr_genre), "Country",
                                     ifelse(grepl("Hip-Hop", orig_data$cdr_genre), "Hip-Hop",
                                             ifelse(grepl("Rock", orig_data$cdr_genre), "Rock",
                                                     ifelse(grepl("Jazz", orig_data$cdr_genre), "Jazz", "O

cleaned_data <- cbind(cleaned_data, bpm=orig_data$bpm, energy=orig_data$energy, danceability = orig_data

#Wind
cleaned_data$has_winds <- ifelse(orig_data$clarinet, "True",
                                ifelse(orig_data$flute_piccolo, "True",
                                        ifelse(orig_data$harmonica, "True",
                                              ifelse(orig_data$kazoo, "True",
                                                    ifelse(orig_data$ocarina, "True",
                                                            ifelse(orig_data$saxophone, "True",
                                                                    ifelse(orig_data$accordion, "True", "False")

#Brass
cleaned_data$has_brass <- ifelse(orig_data$trumpet, "True", "False")

#Percussion
cleaned_data$has_percussion <- ifelse(orig_data$piano_keyboard_based, "True",
                                      ifelse(orig_data$bongos, "True",
                                              ifelse(orig_data$cowbell, "True",
                                                      ifelse(orig_data$handclaps_snaps, "True", "False"))))

#Vocal or Whistle
cleaned_data$has_vocal_or_whistle <- ifelse(orig_data$vocally_based, "True",
                                             ifelse(orig_data$falsetto_vocal, "True",
                                                     ifelse(orig_data$human_whistling, "True", "False")))
```

```

#String
cleaned_data$has_strings <- ifelse(orig_data$bass_based, "True",
                                   ifelse(orig_data$guitar_based, "True",
                                           ifelse(orig_data$orchestral_strings, "True",
                                                   ifelse(orig_data$banjo, "True",
                                                           ifelse(orig_data$mandolin, "True",
                                                                 ifelse(orig_data$pedal_lap_steel, "True",
                                                                 ifelse(orig_data$sitar, "True",
                                                                 ifelse(orig_data$ukulele, "True",
                                                                 ifelse(orig_data$violin, "True", "False")))))))

cleaned_data$year <- as.numeric(substr(orig_data$date, 1, unlist(gregexpr("-", orig_data$date))[1] - 1))

cleaned_data <- cleaned_data[rowSums(is.na(cleaned_data)) == 0 , ]

cutoff <- round(min(cleaned_data$year) + ((max(cleaned_data$year) - min(cleaned_data$year)) / 2))

first_subpop <- cleaned_data[cleaned_data$year < cutoff, ]

second_subpop <- cleaned_data[cleaned_data$year >= cutoff, ]

model_1 <- lm(data = cleaned_data, rating ~ age + race + gender + genre + energy + danceability + happiness)
summary(model_1)

##
## Call:
## lm(formula = rating ~ age + race + gender + genre + energy +
##     danceability + happiness + bpm + loudness_db + length_sec,
##     data = cleaned_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.8024 -1.2244 -0.0203  1.2248  4.5071
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    7.0722615   0.6747344   10.482 < 2e-16 ***
## age           -0.0235027   0.0080080    -2.935  0.003403 **
## raceWhite      -0.4425431   0.1231832    -3.593  0.000341 ***
## genderNot Male  0.2008111   0.1139569     1.762  0.078309 .
## genreJazz       1.1915765   0.8098788     1.471  0.141485
## genreOther      0.2896938   0.3717489     0.779  0.435981
## genrePop       -0.0309389   0.3602833    -0.086  0.931582
## genreRock       0.5806698   0.3666694     1.584  0.113555
## energy         -0.0086781   0.0045002    -1.928  0.054054 .
## danceability   -0.0098823   0.0043489    -2.272  0.023249 *
## happiness       0.0103522   0.0028992     3.571  0.000371 ***
## bpm           -0.0006552   0.0021639    -0.303  0.762097
## loudness_db     0.0442563   0.0223927     1.976  0.048353 *
## length_sec      0.0027224   0.0011510     2.365  0.018188 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
##
## Residual standard error: 1.78 on 1140 degrees of freedom
## Multiple R-squared:  0.05086,    Adjusted R-squared:  0.04003
## F-statistic: 4.699 on 13 and 1140 DF,  p-value: 5.732e-08

model_2 <- lm(data = cleaned_data, rating ~ age + gender + race +age*genre + age*energy + age*danceabil.
summary(model_2)

##
## Call:
## lm(formula = rating ~ age + gender + race + age * genre + age *
##     energy + age * danceability + age * happiness + age * bpm +
##     age * loudness_db + age * length_sec + race * genre + race *
##     energy + race * danceability + race * happiness + race *
##     bpm + race * loudness_db + race * length_sec + +gender *
##     genre + gender * energy + gender * danceability + gender *
##     happiness + gender * bpm + gender * loudness_db + gender *
##     length_sec, data = cleaned_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.7534 -1.1493 -0.0184  1.1834  4.4488
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      9.492e+00  3.315e+00   2.863  0.00427 **
## age             -6.391e-02  9.526e-02  -0.671  0.50243
## genderNot Male    3.706e-01  1.367e+00   0.271  0.78633
## raceWhite        -1.378e+00  1.751e+00  -0.787  0.43120
## genreJazz         4.795e-01  2.932e+00   0.164  0.87014
## genreOther        -1.873e+00  2.329e+00  -0.804  0.42133
## genrePop          -1.092e+00  2.295e+00  -0.476  0.63416
## genreRock         -8.983e-01  2.346e+00  -0.383  0.70184
## energy            1.182e-02  2.016e-02   0.586  0.55775
## danceability      -3.883e-02  2.153e-02  -1.803  0.07165
## happiness         1.270e-02  1.342e-02   0.946  0.34425
## bpm              -3.741e-03  1.020e-02  -0.367  0.71388
## loudness_db       4.239e-02  1.071e-01   0.396  0.69223
## length_sec        1.714e-03  5.632e-03   0.304  0.76091
## age:genreJazz     -2.004e-02  7.403e-02  -0.271  0.78671
## age:genreOther     4.622e-02  6.149e-02   0.752  0.45237
## age:genrePop      -1.034e-02  6.027e-02  -0.172  0.86385
## age:genreRock     -1.651e-03  6.219e-02  -0.027  0.97883
## age:energy        -8.336e-04  6.954e-04  -1.199  0.23091
## age:danceability   9.308e-04  6.870e-04   1.355  0.17574
## age:happiness      1.533e-04  4.417e-04   0.347  0.72865
## age:bpm            7.048e-05  3.299e-04   0.214  0.83087
## age:loudness_db   -1.306e-03  3.605e-03  -0.362  0.71717
## age:length_sec    -4.083e-05  1.896e-04  -0.215  0.82959
## raceWhite:genreJazz 1.658e+00  1.996e+00   0.831  0.40631
## raceWhite:genreOther 9.835e-01  1.363e+00   0.721  0.47084
## raceWhite:genrePop  1.669e+00  1.355e+00   1.232  0.21811
## raceWhite:genreRock 1.991e+00  1.384e+00   1.439  0.15043
## raceWhite:energy    7.483e-04  9.274e-03   0.081  0.93570
## raceWhite:danceability -3.031e-03  8.984e-03  -0.337  0.73587
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## raceWhite:happiness      -7.637e-03  5.873e-03  -1.300  0.19377
## raceWhite:bpm            -3.436e-04  4.386e-03  -0.078  0.93758
## raceWhite:loudness_db    8.519e-02  4.553e-02   1.871  0.06158 .
## raceWhite:length_sec     4.248e-03  2.388e-03   1.778  0.07561 .
## genderNot Male:genreJazz      NA      NA      NA      NA
## genderNot Male:genreOther -1.514e+00  8.832e-01  -1.714  0.08683 .
## genderNot Male:genrePop      -1.077e+00  8.687e-01  -1.240  0.21536
## genderNot Male:genreRock     -1.739e+00  9.057e-01  -1.920  0.05517 .
## genderNot Male:energy        4.152e-03  9.688e-03   0.429  0.66831
## genderNot Male:danceability  1.616e-02  9.138e-03   1.768  0.07734 .
## genderNot Male:happiness     -7.980e-03  6.116e-03  -1.305  0.19225
## genderNot Male:bpm           3.348e-03  4.539e-03   0.738  0.46084
## genderNot Male:loudness_db   -5.966e-03  4.702e-02  -0.127  0.89907
## genderNot Male:length_sec    -3.146e-04  2.412e-03  -0.130  0.89627
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.764 on 1111 degrees of freedom
## Multiple R-squared:  0.09126,    Adjusted R-squared:  0.05691
## F-statistic: 2.657 on 42 and 1111 DF,  p-value: 9.427e-08

model_3 <- lm(data = cleaned_data, rating ~ energy + danceability + happiness + bpm + genre + has_winds +
summary(model_3)

##
## Call:
## lm(formula = rating ~ energy + danceability + happiness + bpm +
##     genre + has_winds + has_vocal_or_whistle + has_strings +
##     has_percussion + has_brass + loudness_db + length_sec, data = cleaned_data)
##
## Residuals:
##    Min     1Q  Median     3Q     Max
## -4.737 -1.236 -0.011  1.225  4.270
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5.7049929  0.6499221   8.778 < 2e-16 ***
## energy         -0.0081115  0.0045427  -1.786  0.07443 .
## danceability   -0.0070830  0.0043716  -1.620  0.10546
## happiness       0.0096091  0.0029278   3.282  0.00106 **
## bpm           -0.0001848  0.0021782  -0.085  0.93242
## genreJazz       1.1333397  0.8346649   1.358  0.17478
## genreOther      0.5824180  0.3703670   1.573  0.11610
## genrePop        0.1440935  0.3657161   0.394  0.69365
## genreRock       0.6023629  0.3706181   1.625  0.10438
## has_windsTrue   0.0362916  0.1833187   0.198  0.84310
## has_vocal_or_whistleTrue 0.5279949  0.1273224   4.147 3.62e-05 ***
## has_stringsTrue  0.1887708  0.1398859   1.349  0.17746
## has_percussionTrue -0.0139132  0.1119683  -0.124  0.90113
## has_brassTrue   0.2101113  0.4356987   0.482  0.62973
## loudness_db     0.0510769  0.0227949   2.241  0.02524 *
## length_sec      0.0023951  0.0011359   2.109  0.03520 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

```

## Residual standard error: 1.787 on 1138 degrees of freedom
## Multiple R-squared:  0.04422,    Adjusted R-squared:  0.03162
## F-statistic: 3.51 on 15 and 1138 DF,  p-value: 6.178e-06

model_4 <- lm(data = cleaned_data, rating ~ has_winds + has_vocal_or_whistle + has_strings + has_percuss
summary(model_4)

##
## Call:
## lm(formula = rating ~ has_winds + has_vocal_or_whistle + has_strings +
##     has_percussion + has_brass + has_winds * genre + has_winds *
##     danceability + has_winds * happiness + has_winds * bpm +
##     has_winds * loudness_db + has_winds * length_sec + has_vocal_or_whistle *
##     genre + has_vocal_or_whistle * energy + has_vocal_or_whistle *
##     danceability + has_vocal_or_whistle * happiness + has_vocal_or_whistle *
##     bpm + has_vocal_or_whistle * loudness_db + has_vocal_or_whistle *
##     length_sec + has_strings * genre + has_strings * energy +
##     has_strings * danceability + has_strings * happiness + has_strings *
##     bpm + has_strings * loudness_db + has_strings * length_sec +
##     has_percussion * genre + has_percussion * energy + has_percussion *
##     danceability + has_percussion * happiness + has_percussion *
##     bpm + has_percussion * loudness_db + has_percussion * length_sec +
##     has_brass * genre + has_brass * energy + has_brass * danceability +
##     has_brass * happiness + has_brass * bpm + has_brass * loudness_db +
##     has_brass * length_sec, data = cleaned_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.997 -1.172  0.000   1.202  4.542
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      5.2087672   2.6966973    1.932  0.0537
## has_windsTrue      1.4552595   2.1642907    0.672  0.5015
## has_vocal_or_whistleTrue -1.9041368   1.8769688   -1.014  0.3106
## has_stringsTrue      1.3070594   2.6169993    0.499  0.6176
## has_percussionTrue  -0.0921369   1.5201446   -0.061  0.9517
## has_brassTrue      -6.2475439   6.9580080   -0.898  0.3694
## genreJazz          0.4922984   2.6893871    0.183  0.8548
## genreOther          0.1793466   1.9804381    0.091  0.9279
## genrePop            0.3456550   1.9960245    0.173  0.8625
## genreRock           1.0968855   2.0234304    0.542  0.5879
## danceability      -0.0035014   0.0125468   -0.279  0.7802
## happiness           0.0055607   0.0081437    0.683  0.4949
## bpm                0.0037523   0.0059431    0.631  0.5279
## loudness_db         0.1032083   0.0712853    1.448  0.1480
## length_sec          0.0048514   0.0034853    1.392  0.1642
## energy             -0.0085751   0.0137528   -0.624  0.5331
## has_windsTrue:genreJazz -5.7941040   4.7874010   -1.210  0.2264
## has_windsTrue:genreOther  0.7202919   1.4517493    0.496  0.6199
## has_windsTrue:genrePop  -0.3833390   1.4012877   -0.274  0.7845
## has_windsTrue:genreRock -0.5559637   1.4039354   -0.396  0.6922
## has_windsTrue:danceability  0.0061141   0.0157568    0.388  0.6981
## has_windsTrue:happiness -0.0139981   0.0093438   -1.498  0.1344
## has_windsTrue:bpm       0.0078633   0.0082521    0.953  0.3409

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## has_windsTrue:loudness_db      0.0530425  0.0641096   0.827   0.4082
## has_windsTrue:length_sec      -0.0054815  0.0039611  -1.384   0.1667
## has_vocal_or_whistleTrue:genreJazz      NA      NA      NA      NA
## has_vocal_or_whistleTrue:genreOther    1.3250590  1.4389365   0.921   0.3573
## has_vocal_or_whistleTrue:genrePop     1.1262175  1.4276689   0.789   0.4304
## has_vocal_or_whistleTrue:genreRock     1.0257745  1.4451321   0.710   0.4780
## has_vocal_or_whistleTrue:energy        0.0103601  0.0111147   0.932   0.3515
## has_vocal_or_whistleTrue:danceability   0.0040007  0.0112948   0.354   0.7233
## has_vocal_or_whistleTrue:happiness     -0.0069374  0.0077023  -0.901   0.3679
## has_vocal_or_whistleTrue:bpm           0.0005392  0.0054469   0.099   0.9212
## has_vocal_or_whistleTrue:loudness_db   -0.0669818  0.0551823  -1.214   0.2251
## has_vocal_or_whistleTrue:length_sec     0.0007063  0.0029255   0.241   0.8093
## has_stringsTrue:genreJazz             7.7275800  4.9908670   1.548   0.1218
## has_stringsTrue:genreOther            0.2539648  1.9769048   0.128   0.8978
## has_stringsTrue:genrePop             -0.2814471  1.9819371  -0.142   0.8871
## has_stringsTrue:genreRock            -0.3318862  2.0133515  -0.165   0.8691
## has_stringsTrue:energy               -0.0090631  0.0126638  -0.716   0.4743
## has_stringsTrue:danceability          -0.0081957  0.0117237  -0.699   0.4847
## has_stringsTrue:happiness             0.0109414  0.0076756   1.425   0.1543
## has_stringsTrue:bpm                  -0.0052299  0.0056603  -0.924   0.3557
## has_stringsTrue:loudness_db          -0.0333618  0.0640447  -0.521   0.6025
## has_stringsTrue:length_sec            -0.0016401  0.0032021  -0.512   0.6086
## has_percussionTrue:genreJazz          -5.5250164  3.9319026  -1.405   0.1603
## has_percussionTrue:genreOther         -0.0466557  1.0748375  -0.043   0.9654
## has_percussionTrue:genrePop          -0.1590962  1.0638089  -0.150   0.8811
## has_percussionTrue:genreRock         -0.4750567  1.0707415  -0.444   0.6574
## has_percussionTrue:energy             0.0094538  0.0095761   0.987   0.3237
## has_percussionTrue:danceability        0.0058585  0.0091888   0.638   0.5239
## has_percussionTrue:happiness          -0.0058776  0.0061890  -0.950   0.3425
## has_percussionTrue:bpm                -0.0008684  0.0045181  -0.192   0.8476
## has_percussionTrue:loudness_db        -0.0208915  0.0486830  -0.429   0.6679
## has_percussionTrue:length_sec          -0.0017813  0.0024352  -0.731   0.4646
## has_brassTrue:genreJazz               -4.6713526  3.5974616  -1.299   0.1944
## has_brassTrue:genreOther              -0.6998014  4.4527039  -0.157   0.8751
## has_brassTrue:genrePop                1.5327089  2.8874160   0.531   0.5956
## has_brassTrue:genreRock              -0.4389169  4.7225972  -0.093   0.9260
## has_brassTrue:energy                  0.0501310  0.1065643   0.470   0.6381
## has_brassTrue:danceability             0.0070456  0.0539792   0.131   0.8962
## has_brassTrue:happiness               0.0440398  0.0462074   0.953   0.3408
## has_brassTrue:bpm                     -0.0255653  0.0431352  -0.593   0.5535
## has_brassTrue:loudness_db             -0.1865797  0.3569560  -0.523   0.6013
## has_brassTrue:length_sec               0.0062546  0.0201997   0.310   0.7569
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## Residual standard error: 1.793 on 1090 degrees of freedom
```

```
## Multiple R-squared:  0.07865,    Adjusted R-squared:  0.02539
```

```
## F-statistic: 1.477 on 63 and 1090 DF,  p-value: 0.01049
```

```
model_5 <- lm(data = cleaned_data, rating ~ age + gender + race + energy + danceability + happiness + 1)
summary(model_5)
```

```
##
```

```
## Call:
```

```
## lm(formula = rating ~ age + gender + race + energy + danceability +
```

```

##      happiness + bpm + genre + has_winds + has_vocal_or_whistle +
##      has_strings + has_percussion + has_brass + loudness_db +
##      length_sec, data = cleaned_data)
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -4.6570 -1.2250 -0.0168  1.2266  4.5164
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      6.607992   0.684180   9.658 < 2e-16 ***
## age             -0.021643   0.008005  -2.704 0.006962 **
## genderNot Male    0.224080   0.113358   1.977 0.048313 *
## raceWhite        -0.449231   0.122483  -3.668 0.000256 ***
## energy           -0.007580   0.004498  -1.685 0.092240 .
## danceability     -0.009047   0.004347  -2.081 0.037643 *
## happiness         0.009635   0.002900   3.322 0.000923 ***
## bpm              -0.000200   0.002155  -0.093 0.926070
## genreJazz         1.152965   0.830662   1.388 0.165407
## genreOther        0.259902   0.373874   0.695 0.487099
## genrePop          -0.066628   0.364021  -0.183 0.854804
## genreRock         0.569321   0.367789   1.548 0.121912
## has_windsTrue     0.034225   0.181546   0.189 0.850503
## has_vocal_or_whistleTrue 0.511749   0.126451   4.047 5.54e-05 ***
## has_stringsTrue   0.225167   0.138664   1.624 0.104690
## has_percussionTrue -0.007714   0.110786  -0.070 0.944501
## has_brassTrue     0.253570   0.431093   0.588 0.556514
## loudness_db       0.043779   0.022611   1.936 0.053092 .
## length_sec        0.002733   0.001148   2.379 0.017508 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.768 on 1135 degrees of freedom
## Multiple R-squared:  0.06724,    Adjusted R-squared:  0.05245
## F-statistic: 4.545 on 18 and 1135 DF,  p-value: 1.025e-09
model_6 <- lm(data = cleaned_data, rating ~ age + race + gender + has_winds + has_vocal_or_whistle + has_strings + has_percussion + has_brass + loudness_db + length_sec)
summary(model_6)

##
## Call:
## lm(formula = rating ~ age + race + gender + has_winds + has_vocal_or_whistle +
##      has_strings + has_percussion + has_brass + has_winds * genre +
##      has_winds * danceability + has_winds * happiness + has_winds *
##      bpm + has_winds * loudness_db + has_winds * length_sec +
##      has_vocal_or_whistle * genre + has_vocal_or_whistle * energy +
##      has_vocal_or_whistle * danceability + has_vocal_or_whistle *
##      happiness + has_vocal_or_whistle * bpm + has_vocal_or_whistle *
##      loudness_db + has_vocal_or_whistle * length_sec + has_strings *
##      genre + has_strings * energy + has_strings * danceability +
##      has_strings * happiness + has_strings * bpm + has_strings *
##      loudness_db + has_strings * length_sec + has_percussion *
##      genre + has_percussion * energy + has_percussion * danceability +
##      has_percussion * happiness + has_percussion * bpm + has_percussion *
##      loudness_db + has_percussion * length_sec + has_brass * genre +

```

```

##      has_brass * energy + has_brass * danceability + has_brass *
##      happiness + has_brass * bpm + has_brass * loudness_db + has_brass *
##      length_sec + age * genre + age * energy + age * danceability +
##      age * happiness + age * bpm + age * loudness_db + age * length_sec +
##      race * genre + race * energy + race * danceability + race *
##      happiness + race * bpm + race * loudness_db + race * length_sec +
##      +gender * genre + gender * energy + gender * danceability +
##      gender * happiness + gender * bpm + gender * loudness_db +
##      gender * length_sec, data = cleaned_data)
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -4.9954 -1.0902 -0.0149  1.2330  4.6043
##
## Coefficients: (3 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.007e+01  4.475e+00   2.251  0.0246 *
## age              -9.384e-02  1.020e-01  -0.920  0.3579
## raceWhite        -2.137e+00  1.807e+00  -1.183  0.2372
## genderNot Male    4.438e-01  1.447e+00   0.307  0.7592
## has_windsTrue     1.656e+00  2.192e+00   0.755  0.4503
## has_vocal_or_whistleTrue -2.235e+00  1.943e+00  -1.150  0.2503
## has_stringsTrue   9.370e-01  2.656e+00   0.353  0.7243
## has_percussionTrue 4.253e-01  1.528e+00   0.278  0.7808
## has_brassTrue     -3.920e+00  7.276e+00  -0.539  0.5902
## genreJazz         -2.881e+00  1.193e+01  -0.241  0.8093
## genreOther        -3.139e+00  3.371e+00  -0.931  0.3521
## genrePop          -2.175e+00  3.363e+00  -0.647  0.5179
## genreRock         -1.313e+00  3.426e+00  -0.383  0.7016
## danceability      -3.405e-02  2.452e-02  -1.388  0.1653
## happiness         8.036e-03  1.536e-02   0.523  0.6008
## bpm              -1.859e-03  1.154e-02  -0.161  0.8721
## loudness_db       9.088e-02  1.278e-01   0.711  0.4772
## length_sec        3.448e-03  6.644e-03   0.519  0.6039
## energy            1.164e-02  2.444e-02   0.476  0.6340
## has_windsTrue:genreJazz -5.491e+00  9.280e+00  -0.592  0.5541
## has_windsTrue:genreOther -5.961e-02  1.478e+00  -0.040  0.9678
## has_windsTrue:genrePop -1.222e+00  1.427e+00  -0.857  0.3918
## has_windsTrue:genreRock -1.323e+00  1.430e+00  -0.925  0.3552
## has_windsTrue:danceability 7.029e-03  1.586e-02   0.443  0.6577
## has_windsTrue:happiness -9.529e-03  9.369e-03  -1.017  0.3094
## has_windsTrue:bpm    6.997e-03  8.289e-03   0.844  0.3988
## has_windsTrue:loudness_db 4.660e-02  6.376e-02   0.731  0.4650
## has_windsTrue:length_sec -4.291e-03  4.025e-03  -1.066  0.2866
## has_vocal_or_whistleTrue:genreJazz      NA         NA         NA         NA
## has_vocal_or_whistleTrue:genreOther    1.367e+00  1.527e+00   0.895  0.3710
## has_vocal_or_whistleTrue:genrePop    1.101e+00  1.517e+00   0.726  0.4680
## has_vocal_or_whistleTrue:genreRock    1.052e+00  1.533e+00   0.686  0.4928
## has_vocal_or_whistleTrue:energy       7.535e-03  1.107e-02   0.681  0.4960
## has_vocal_or_whistleTrue:danceability 4.324e-03  1.127e-02   0.384  0.7013
## has_vocal_or_whistleTrue:happiness    -4.392e-03  7.780e-03  -0.565  0.5725
## has_vocal_or_whistleTrue:bpm         2.609e-03  5.459e-03   0.478  0.6328
## has_vocal_or_whistleTrue:loudness_db  -5.892e-02  5.554e-02  -1.061  0.2890
## has_vocal_or_whistleTrue:length_sec    1.291e-03  2.929e-03   0.441  0.6596

```

## has_stringsTrue:genreJazz	6.576e+00	5.019e+00	1.310	0.1904
## has_stringsTrue:genreOther	1.076e+00	2.018e+00	0.533	0.5942
## has_stringsTrue:genrePop	6.139e-01	2.026e+00	0.303	0.7619
## has_stringsTrue:genreRock	4.351e-01	2.060e+00	0.211	0.8327
## has_stringsTrue:energy	-1.078e-02	1.270e-02	-0.849	0.3964
## has_stringsTrue:danceability	-8.805e-03	1.174e-02	-0.750	0.4535
## has_stringsTrue:happiness	1.175e-02	7.702e-03	1.526	0.1274
## has_stringsTrue:bpm	-7.624e-03	5.698e-03	-1.338	0.1812
## has_stringsTrue:loudness_db	-2.987e-02	6.370e-02	-0.469	0.6392
## has_stringsTrue:length_sec	-1.985e-03	3.195e-03	-0.621	0.5345
## has_percussionTrue:genreJazz	-2.622e+00	7.394e+00	-0.355	0.7229
## has_percussionTrue:genreOther	-1.469e-01	1.085e+00	-0.135	0.8924
## has_percussionTrue:genrePop	-3.499e-01	1.076e+00	-0.325	0.7452
## has_percussionTrue:genreRock	-5.690e-01	1.082e+00	-0.526	0.5992
## has_percussionTrue:energy	7.576e-03	9.632e-03	0.787	0.4317
## has_percussionTrue:danceability	3.699e-03	9.207e-03	0.402	0.6880
## has_percussionTrue:happiness	-5.021e-03	6.267e-03	-0.801	0.4232
## has_percussionTrue:bpm	-8.827e-04	4.512e-03	-0.196	0.8449
## has_percussionTrue:loudness_db	1.382e-02	4.894e-02	0.282	0.7777
## has_percussionTrue:length_sec	-1.430e-03	2.441e-03	-0.586	0.5583
## has_brassTrue:genreJazz	-5.004e+00	4.652e+00	-1.076	0.2823
## has_brassTrue:genreOther	1.738e+00	4.644e+00	0.374	0.7084
## has_brassTrue:genrePop	3.207e+00	3.057e+00	1.049	0.2944
## has_brassTrue:genreRock	2.386e+00	4.848e+00	0.492	0.6227
## has_brassTrue:energy	3.060e-03	1.064e-01	0.029	0.9771
## has_brassTrue:danceability	2.427e-03	5.526e-02	0.044	0.9650
## has_brassTrue:happiness	5.678e-02	4.555e-02	1.247	0.2129
## has_brassTrue:bpm	-2.731e-02	4.325e-02	-0.631	0.5280
## has_brassTrue:loudness_db	-8.992e-02	3.959e-01	-0.227	0.8204
## has_brassTrue:length_sec	1.086e-03	2.056e-02	0.053	0.9579
## age:genreJazz	8.551e-02	2.281e-01	0.375	0.7078
## age:genreOther	4.948e-02	6.768e-02	0.731	0.4649
## age:genrePop	9.329e-03	6.645e-02	0.140	0.8884
## age:genreRock	6.583e-03	6.825e-02	0.096	0.9232
## age:energy	-6.738e-04	7.210e-04	-0.934	0.3503
## age:danceability	1.047e-03	7.099e-04	1.474	0.1407
## age:happiness	5.111e-05	4.537e-04	0.113	0.9103
## age:bpm	1.274e-04	3.456e-04	0.369	0.7125
## age:loudness_db	-2.011e-03	3.725e-03	-0.540	0.5895
## age:length_sec	-5.551e-05	1.964e-04	-0.283	0.7775
## raceWhite:genreJazz	NA	NA	NA	NA
## raceWhite:genreOther	1.055e+00	1.399e+00	0.754	0.4508
## raceWhite:genrePop	1.774e+00	1.391e+00	1.276	0.2023
## raceWhite:genreRock	2.065e+00	1.420e+00	1.454	0.1463
## raceWhite:energy	2.937e-03	9.608e-03	0.306	0.7599
## raceWhite:danceability	-4.857e-03	9.259e-03	-0.525	0.6000
## raceWhite:happiness	-7.885e-03	6.200e-03	-1.272	0.2037
## raceWhite:bpm	2.478e-03	4.533e-03	0.547	0.5847
## raceWhite:loudness_db	7.564e-02	4.668e-02	1.620	0.1055
## raceWhite:length_sec	5.382e-03	2.522e-03	2.134	0.0331 *
## genderNot Male:genreJazz	NA	NA	NA	NA
## genderNot Male:genreOther	-1.803e+00	9.657e-01	-1.867	0.0621 .
## genderNot Male:genrePop	-1.421e+00	9.543e-01	-1.489	0.1368
## genderNot Male:genreRock	-2.002e+00	9.873e-01	-2.028	0.0428 *

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## genderNot Male:energy          2.937e-03  9.886e-03   0.297   0.7665
## genderNot Male:danceability    1.245e-02  9.364e-03   1.329   0.1840
## genderNot Male:happiness       -5.785e-03  6.280e-03  -0.921   0.3572
## genderNot Male:bpm             5.295e-03  4.664e-03   1.135   0.2565
## genderNot Male:loudness_db     -9.867e-03  4.799e-02  -0.206   0.8371
## genderNot Male:length_sec      6.293e-04  2.493e-03   0.252   0.8007
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.758 on 1059 degrees of freedom
## Multiple R-squared:  0.1395, Adjusted R-squared:  0.06315
## F-statistic: 1.827 on 94 and 1059 DF, p-value: 6.866e-06

```